Prakhar **Verma**

Machine Learning Researcher · PhD

Helsinki, Finland.

prakhar.verma7@gmail.com | www.prakharverma.github.io | vermaprakhar | Prakhar Verma

Research Summary

Machine learning researcher specialising in probabilistic modeling, Bayesian inference, and large language models, with experience spanning Microsoft Research, Adobe Research, the University of Oxford, and production-scale LLM reasoning systems at Inven. My work focuses on uncertainty-aware reasoning, sequential decision-making, and scalable inference for language models and probabilistic systems. Recent research includes efficient test-time planning for retrieval-augmented generation (RAG) and Bayesian causal discovery with language model priors. I develop theoretically grounded methods with practical impact across both research and production settings.

Experience

Inven
Helsinki, Finland

Senior Machine Learning Engineer — *October 2025 - Present*

- Developed production-scale **LLM systems for entity understanding**, transforming noisy web data, official documents, and business registries into structured relational representations across **millions of companies**.
- Designed **hybrid semantic retrieval and information extraction systems** for large-scale entity resolution and relationship discovery over heterogeneous business data sources.
- Developed machine learning approaches for **legal entity matching** and **ownership structure modeling** under noisy and incomplete observations, enabling robust grounding of real-world entities.

Adobe Research
Bangalore, India

Research Intern — *June 2024 - August 2024*

- Developed a **Bayesian framework for sequential causal discovery** that integrates language model priors with observational data for uncertainty-aware structure learning.
- Proposed an iterative causal refinement approach to mitigate **dual-bias** arising from both language model priors and observational data.
- Led to a patent-pending method and a publication at TMLR.

Microsoft Research
Bangalore, India

Research Intern — *March 2024 - May 2024*

- Developed **Plan*RAG**, a framework for multi-hop **retrieval-augmented generation (RAG)** that builds **traceable reasoning DAGs** for efficient retrieval and inference.
- Reduced inference latency and retrieval cost through **efficient test-time planning** strategies.
- Led to a publication at the ICLR 2025 Workshop on Reasoning and Planning for LLMs.

University of Oxford
Oxford, United Kingdom

Visiting Researcher — *July 2023 - September 2023*

- Collaborated with Prof. Seth Flaxman and Elizaveta Semenova on **encoding prior knowledge** into probabilistic models for life science applications.
- Developed **scalable inference techniques** for Bayesian generative models in epidemiology and medical applications.
- Led to a research publication on scalable Bayesian deep generative models.

SpectacularAI
Espoo, Finland

Research Engineer — *September 2021 - September 2022*

- Developed **uncertainty estimation** methods for deep learning models using approximate Bayesian inference techniques, including MC Dropout and Laplace approximation, to improve model robustness under uncertainty.

TomTom
Pune, India

Software Engineer (R&D) — *July 2016 - August 2019*

- Developed a **semantic segmentation** pipeline for extracting map features from satellite imagery and integrating them into large-scale mapping databases.

Selected Publications

- **Prakhar Verma**, David Arbour, Sunav Choudhary, Harshita Chopra, Arno Solin, Atanu R. Sinha. Sequential Causal Discovery with Noisy Language Model Priors. *Transactions on Machine Learning Research (TMLR), 2026.*
- **Prakhar Verma**, Sukruta Prakash Midigeshi, Gaurav Sinha, Arno Solin, Nagarajan Natarajan, Amit Sharma. Plan*RAG: Efficient Test-Time Planning for Retrieval Augmented Generation. Workshop on *Reasoning and Planning for Large Language Models* at ICLR 2025.
- **Prakhar Verma**, Vincent Adam, Arno Solin. Variational Gaussian Process Diffusion Processes. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.*
- Paul Edmund Chang*, **Prakhar Verma***, S.T. John, Arno Solin, Mohammad Emtiyaz Khan. Memory-based dual Gaussian processes for sequential learning. *International Conference on Machine Learning (ICML), 2023. (Oral Presentation)*
- Arno Solin, Ella Tamir, **Prakhar Verma**. Scalable Inference in SDEs by Direct Matching of the Fokker–Planck–Kolmogorov Equation. *Advances in Neural Information Processing Systems 35* (NeurIPS), 2021.
- **Prakhar Verma**, Paul Chang, Arno Solin, Mohammad Emtiyaz Khan. Sequential Learning in GPs with Memory and Bayesian Leverage Score. *Asian Conference in Machine Learning (ACML) workshop "Continual Lifelong Learning" 2022 (Contributed talk)*.
- Elizaveta Semenova, **Prakhar Verma**, Max Cairney-Leeming, Arno Solin, Samir Bhatt, Seth Flaxman. PriorCVAE: Scalable MCMC parameter inference with Bayesian deep generative modelling.

Education

Aalto University — *Finland*

Doctor of Philosophy (Ph.D.) — *2022 - June 2026*

- Research on **probabilistic machine learning**, **Bayesian inference**, and **uncertainty quantification** with applications to sequential decision-making and large language models (LLMs). Supervised by Prof. Arno Solin.
- **Thesis:** Scalable Probabilistic Inference for Sequential Stochastic Models

Aalto University — *Finland*

Master of Science (M.Sc.) - 4.7/5 (Pass with Honors) — *2019 - 2021*

- Major in Machine Learning, Data Science, and Artificial Intelligence and Minor in Mathematics
- **Thesis:** *Sparse Gaussian Processes for Stochastic Differential Equations*

Uttarakhand Technical University — *India*

Bachelor of Technology (B.Tech.) - 79.03% (First Division with Honors) — *2012 - 2016*

- Specialization in Information Technology
- **Thesis:** *Development of Automated GIS Tools on Various Platforms* with TomTom, India.

Research Interests & Technical Skills

- **Research Interests:** Probabilistic Machine Learning, Bayesian Inference, Large Language Models, Retrieval-Augmented Generation, Sequential Decision-Making, Generative Modeling, Causal Discovery, Uncertainty Quantification
- **Frameworks & Tools:** PyTorch, JAX, Hugging Face Transformers, TensorFlow, GPflow, PySpark, Docker, AWS, Weights & Biases, Hydra
- **Programming Languages:** Python, Java, C

Academic Service

- Workflow Chair, International Conference on Artificial Intelligence and Statistics (AISTATS), 2026
- Regular reviewer for NeurIPS, AISTATS, ICLR, ICML, CoLLAs

Selected Honors

- Awarded *"Nokia Scholarship"* in *2024* for exceptional progress and research excellence during doctoral studies.
- Awarded *"Dean Scholarship"* in *2020* and *2021* at Aalto University for commendable academic progress during MSc studies.
- Winner of TomTom *"Innovation Day 2018"*, presented an AI plugin which bridges the gap between machines and cartographers.